

2023



SORACOM Achieves Democratization of IoT Technology with an IoT Solution on AWS, Available in over 160 Countries and Regions

Learn how Soracom in the information and communication industry democratized IoT technology while reducing costs using AWS.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Architecture](#) | [Outcome](#) | [AWS Services Used](#)

5 million subscribers

in 7 years after launch

160 countries/regions

reached globally

20,000 customers

20% reduction

of AWS cost through AWS Enterprise S



Hi, I can connect you with an AWS representative or answer questions you have on AWS.



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



Overview

[Soracom](#), which provides the Internet of Things (IoT) solution SORACOM, built and has operated its core communication and backend functions on Amazon Web Services (AWS) since its launch in 2015. The number of



subscribers of SORACOM Air, a wireless IoT communication service using cellular networks, has exceeded 5 million in 7 years. SORACOM's services have expanded to 160 countries and regions around the world. They are used for a wide range of applications, such as remote monitoring and payment, social infrastructure, factory visualization, and agriculture and fishing.



Opportunity | Implementing Functions Necessary for IoT as Software on the Cloud

As a technology that connects devices, such as sensors, to the cloud through the internet, IoT is attracting attention in promoting digital transformation. SORACOM implements the functions necessary to realize the IoT as software on the cloud, making it easy to implement.

“With the idea of democratizing technology, we launched SORACOM to realize a solution that virtually anyone could easily use to build IoT systems quickly, at a low initial cost. Since we released SORACOM Air in 2015, we have been providing an innovative platform service,” says Akio Katayama, senior executive officer and senior vice president of engineering at Soracom.

Solution | Composing Functional Microservice Components

Soracom has been operating all its services on AWS since releasing SORACOM Air in 2015 because AWS is a global infrastructure available all over the world. AWS provides scalability to facilitate rapid expansion of transactions, and it is robust enough to support the company's communication platforms. "SORACOM is a platform designed to be based on AWS. Using AWS to span to regions around the world and provide APIs to control various functions, we can easily expand our services outside of Japan. Considering our platform serves billions or tens of billions of IoT device connections, the scalability AWS provides is essential. It was also attractive that AWS has all the necessary services for security and fault tolerance," says Katayama.

The communication infrastructure of SORACOM consists of Polaris—which controls communication such as packet transmission, line management, bandwidth control, and access control—and Dipper, which performs backend processing such as session management, authentication, and billing. Both are loosely connected and linked with functional microservice components. Polaris is a stateless communication function, making it easy to fail over. The Dipper backend oversees the stateful area, and Dipper itself consists of microservices. It uses [Amazon DynamoDB](#) as the data store, creating robustness by its high stability and virtually unlimited capacity. The microservices are monitored by the monitoring system Hubble and can recover automatically in the event of error.

"We designed the architecture with fault tolerance in mind and in line with AWS best practices. We introduced microservices to achieve redundancy using multiple Availability Zones, making it easier to update services with independent lifecycles and achieve development, maintenance, and operation with a small number of people," says Katayama.

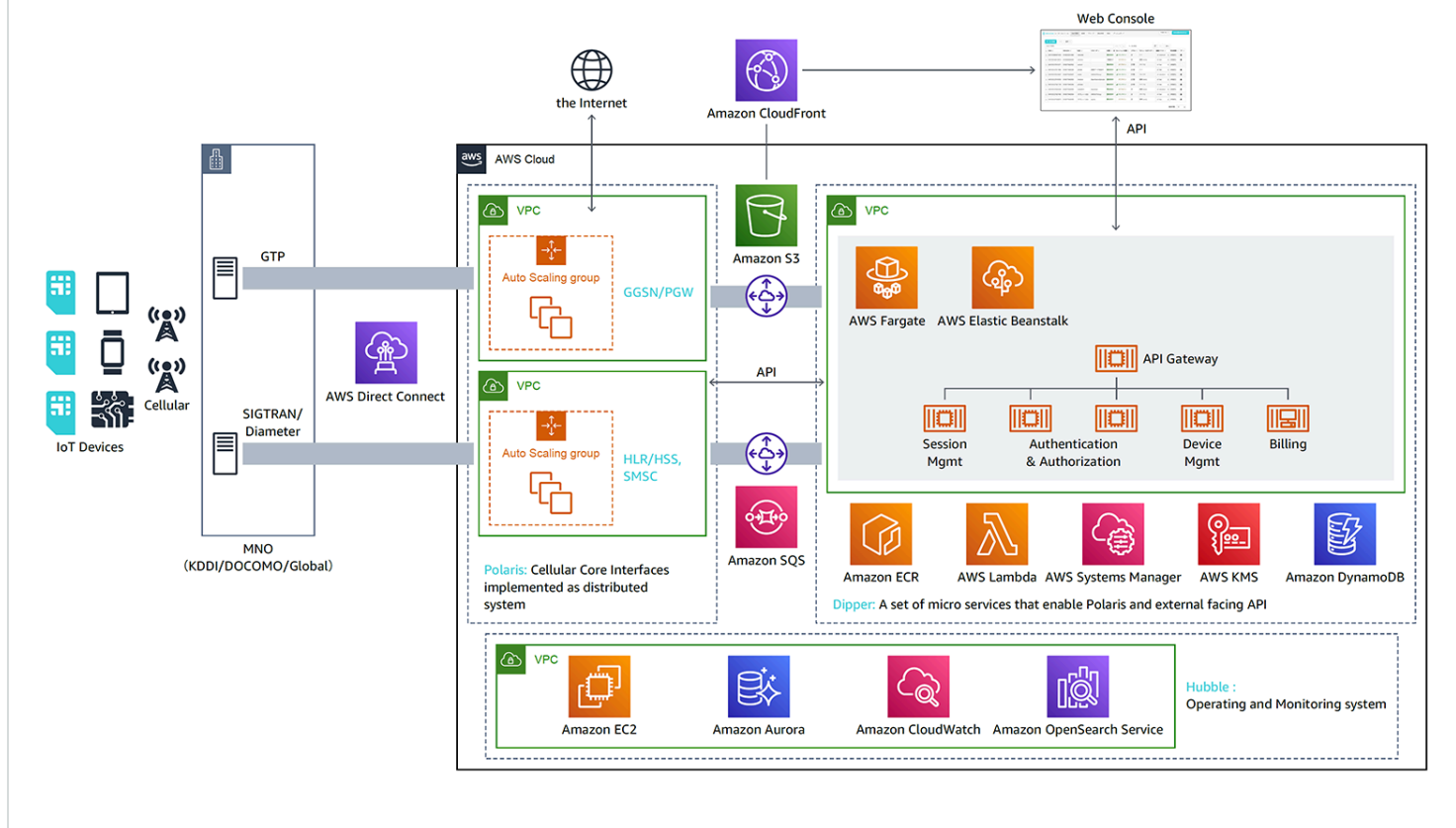
The company began development of SORACOM in the spring of 2015. Only 6 months later, in September, it released SORACOM Air and SORACOM Beam, a data transfer support service. At the time, 10 members of the developer team were assigned to developing packet-exchange functions, billing and authentication functions, and operation management functions. They continued development while having biweekly meetings. "Our AWS contact gave us advice about network-related processing, such as assigning IP addresses and connecting to a carrier, which was very helpful," says Katayama.

The team has continued to enhance functionality since the solution's release. It currently has 40 engineers divided into 7 teams, who provide releases on a biweekly cycle while developing new services. They also modify the architecture continuously, such as by migrating services to containers. In 2021, the company engaged [AWS Enterprise Support](#). Soracom received technical and cost-optimization support and introduction to new services.

"Through AWS Enterprise Support cost-optimization support, we applied [Amazon Elastic Compute Cloud](#) (Amazon EC2) [Reserved Instances](#) and [Savings Plans](#) and ceased unnecessary services. As a result, we reduced our AWS costs by approximately 20 percent over a few months despite the steady increase in the number of subscribers. Using blue/green deployment on AWS, we were able to automate deployment," says Katayama.

Architecture





Outcome | Using 20,000 Implementations Worldwide for a Wide Range of Applications

The SORACOM services grew significantly in the 7 years after its launch, and as of December 2022, the services were provided in 160 countries and regions around the world. SORACOM Air supports cellular communications (5G/LTE/LTE-M) that can be used with domestic and international communication carriers to Sigfox and LoRaWAN for IoT. Its subscribers exceeded 5 million. The total number of customers worldwide was around 20,000. The solution's wide range of applications include dynamic management that captured location information of goods and people, remote monitoring that achieved equipment failure detection and safety maintenance, payment terminals with an eye on the cashless era, factory visualization that captures the operating status of lines, social infrastructure like automatic meter reading and disaster prevention for gas companies, and agriculture, fishery, and livestock farming for primary industries.

SORACOM services have been increasingly incorporated into consumer products and services, and commercialized interpretation devices that can be used all over the world. "Many customers, from Japan as well as overseas, are using SORACOM services. In Japan, IoT is making its way into local areas. Our user base is expanding. Our services are used overseas for purposes we never anticipated, such as supporting the health management and pollination activities of honeybees," says Katayama.

SORACOM's services continue to evolve, and new services continue to be released. In May 2022, the company released SORACOM Cloud Camera Services in Japan, which can manage multiple surveillance camera devices from a smartphone app. In July 2022, message transmission functions using artificial satellites were integrated into the solution, and further coverage expansion is underway.

"SORACOM is a global solution," says Katayama. "Even if the name SORACOM doesn't stand out, we will continue to increase the number of customers who formalize new business using IoT. Going forward, the company will continue



to contribute to social and business innovation while working to democratize IoT. We look forward to continued technical and business support from AWS.”

About Soracom

The company began its service in 2015 with the vision of “connecting people and things around the world and creating a society that resonates.” It provides the IoT platform SORACOM with the mission of democratizing technology. The company’s service portfolio spans IoT communication led by SORACOM Air, devices, networks, applications, and interfaces.



Akio Katayama

AWS Services Used

Amazon DynamoDB

Amazon DynamoDB is a serverless, NoSQL, fully managed database service with single-digit millisecond response times at any scale, enabling you to develop and run modern applications while only paying for what you use.

[Learn more »](#)



AWS Enterprise Support

AWS Enterprise Support provides a comprehensive suite of resources, including proactive planning, advisory services, automation tools, communication channels, and 24/7 expert support.

[Learn more »](#)

Savings Plans

Savings Plans is a flexible pricing model that can help you reduce your bill by up to 72% compared to On-Demand prices, in exchange for a one- or three-year hourly spend commitment.

[Learn more »](#)

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)

The above content is based on the interview on November, 2022

Get Started

Organizations of all sizes across all industries are transforming their businesses and delivering on their missions every day using AWS. Contact our experts and start your own AWS journey today.

[Contact Sales](#)

Resources for AWS

[Getting Started](#)

Developers on AWS

[Developer Center](#)



[Contact Us](#)

Learn About AWS

- What Is AWS?
- What Is Cloud Computing?
- AWS Accessibility
- AWS Inclusion, Diversity & Equity
- What Is DevOps?
- What Is a Container?
- What Is a Data Lake?
- What is Artificial Intelligence (AI)?
- What is Generative AI?
- What is Machine Learning (ML)?
- AWS Cloud Security
- What's New
- Blogs
- Press Releases

- Training and Certification
- AWS Solutions Library
- Architecture Center
- Product and Technical FAQs
- Analyst Reports
- AWS Partners
- SDKs & Tools
- .NET on AWS
- Python on AWS
- Java on AWS
- PHP on AWS
- JavaScript on AWS

- Get Expert Help
- File a Support Ticket
- AWS re:Post
- Knowledge Center
- AWS Support Overview
- Legal
- AWS Careers



Amazon is an Equal Opportunity Employer: *Minority / Women / Disability / Veteran / Gender Identity / Sexual Orientation / Age.*

Language

- عربي |
- Bahasa Indonesia |
- Deutsch |
- English |
- Español |
- Français |
- Italiano |
- Português |
- Tiếng Việt |
- Türkçe |
- Русский |
- ไทย |
- 日本語 |
- 한국어 |
- 中文 (简体) |
- 中文 (繁體)

